

SMTH 322--Tutorial 1

- Perform each of the indicated operations:
 - $(-1 + 2i)\{(7 - 5i) + (-3 + 4i)\}$
 - $\frac{5+5i}{3-4i} + \frac{20}{4+3i}$
 - $\frac{3i^{30}-i^{19}}{2i-1}$
- Suppose $z_1 = 2 + i$, $z_2 = 3 - 2i$ and $z_3 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$. Evaluate each of the following
 - $(\bar{z}_3)^4$
 - $\left| \frac{2z_2+z_1-5-i}{2z_1-z_2+3-i} \right|^2$
- Prove that
 - $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$
 - $|z_1 z_2| = |z_1| |z_2|$
 - $|z_1 + z_2| \leq |z_1| + |z_2|$
- Use the definition of complex number as an ordered pair of real numbers to prove that $(a, b) = a(1, 0) + b(0, 1)$ where $(0, 1)(0, 1) = (-1, 0)$
- Suppose $z_1 = (a_1, b_1)$, $z_2 = (a_2, b_2)$ and $z_3 = (a_3, b_3)$. Prove the distributive law $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$
- Express each of the following complex numbers in polar form
 - $2 + 2\sqrt{3}i$
 - $-\sqrt{6} - \sqrt{2}i$
- Suppose $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$. Prove:
 - $z_1 z_2 = r_1 r_2 \{\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)\}$
 - $\frac{z_1}{z_2} = \frac{r_1}{r_2} \{\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)\}$
- Prove De Moivre's theorem: $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$ where n is any positive integer.
- Prove the identities:
 - $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$
 - $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$
 - $\sin^3 \theta = \frac{3}{4} \sin \theta - \frac{1}{4} \sin 3\theta$
- Find all values of z for which $z^3 = -32$
- Find the indicated roots of $(-2\sqrt{3} - 2i)^{1/4}$
- Find all the 5th root of unity
- Suppose $n = 2, 3, 4, \dots$ Prove that

$$\cos \frac{2\pi}{n} + \cos \frac{4\pi}{n} + \cos \frac{6\pi}{n} + \dots + \cos \frac{2(n-1)\pi}{n} = -1$$